

A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 18

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• MEDCs: market demand and higher prices for the flavour of carnivorous fish (salmon, trout) justify high inputs (fishmeal, cooler climates suit them) • LEDCs: herbivores / omnivores feed on natural vegetation, phytoplankton and wastes so need low inputs; warm climates suit tilapia • Herbivores are more energy-efficient (lower trophic level) so more food per unit area	3
01.2	• Some genetically female (XX) fish are treated with testosterone so they become functional males producing only X-carrying sperm • Crossed with normal females (XX) all offspring are XX females • Female trout meat has a better flavour; in tilapia the reverse (all males) because males grow larger / more efficiently	3
01.3	• Salmon need high oxygen: water sprays / weirs aerate; wastes removed to prevent deoxygenation • Constant circular flow keeps fish swimming the same way - fewer collisions / injuries, more muscular fish, higher stocking density • Artificial lighting delays maturation (salmon stop growing at maturity) so larger fish; controls smoltification • Cool sites (Scotland, Norway) because warm water holds less oxygen although warmth increases growth	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 352-355; AQA spec 3.5.2.3.	

Question	Marking guidance	Marks
02.1	• Species: a group of closely related organisms that naturally interbreed to produce fertile offspring and are reproductively isolated from other groups (Passer domesticus, house sparrow); the only 'real' taxon • Niche: the role a species plays in its habitat - how it uses resources and relates to other species (tawny owl: nocturnal carnivore nesting in tree holes eating small mammals); two species cannot share exactly the same niche in one habitat • Community: all the populations of all species (plants, animals, fungi, bacteria, protists, archaea) living in an area • Biome: a large geographical region with a specific climate and characteristic community, including all areas where it occurs (all tropical rainforests, tundra, savannah)	8
02.2	• An ecosystem is the biotic community plus abiotic factors of one area, relatively self-contained (a particular coral reef) • A biome includes all separate ecosystems with the same climate and community type (all coral reefs); each reef is a different ecosystem within the same biome	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 82-85; AQA spec 3.1.3.2.	
03.1	• $1000 / 100\ 000 \times 100 = 1\%$	1
03.2	• Some light is reflected by the leaves • Some light is transmitted (passes through the leaves without being absorbed) or is of wavelengths not absorbed • Much of the energy captured is used in the plant's metabolic processes and released as low energy-density heat (respiration)	3
03.3	• Less energy is passed to each trophic level than was received from the previous one (heat losses in respiration) • After four levels too little energy remains to support a further population	2
03.4	• Permanent grassland is a more sustainable land use than arable crops in these areas (soil / rainfall unsuitable for crops) • Grazing animals can digest grass cellulose using symbiotic bacteria in their stomachs, producing food that humans can eat	2
03.5	• Pigs are omnivores that can eat food wastes not wanted by people (eg cheese-making wastes, catering leftovers) • So edible meat is produced from material that would otherwise be wasted, without using land / crops that could feed people	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 326-328; AQA spec 3.5.1.2.	
04.1	• A: coppice rotation creates a patchwork of different ages / light levels - clearings with wildflowers for butterflies, dense regrowth for nesting birds; oak supports many invertebrates • B: same tree species but the closed canopy shades the floor so few flowering plants, fewer invertebrates and butterflies; fewer niches • C: non-indigenous single-species, single-age conifers support few native invertebrates; dense shade; no clearings or dead wood; needles acidify soil • Habitat diversity / structural variety increases the range of niches and hence species	5

Question	Marking guidance	Marks
04.2	<i>Any 3 from:</i> • Designated protected areas: SSSIs, NNRs (Sherwood Forest), SACs (Epping Forest), National Parks • Legal protection of ancient woodland: planning permission refused for developments destroying irreplaceable habitats (2012 NPPF) • Conservation management: coppicing, creating clearings, planting mixed native species, leaving dead wood, controlling deer • Planting new native woodlands linking fragments (corridors); Woodland Trust / National Trust ownership	3
04.3	• No species is completely dominant and food webs have many alternative links • So a change in the population of one species has a relatively small impact on the whole community	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 50-52; AQA spec 3.1.2.3.4.	
05.1	• Tracking: continuous data on movements, territory size, migration routes and habitat use without disturbance, over large areas and long periods; acoustic tags in seawater; geolocators for small birds; more representative than occasional sightings • DNA: identifies individuals, gene pool size / inbreeding and regional populations (tracing ivory, timber, fish); eDNA detects rare or hidden aquatic species (great crested newt, invasive carp) from shed cells without capture • Image recognition: software identifies individuals from unique markings (tiger stripes, whale shark spots, fin damage) in camera-trap / drone / satellite images, giving population size, lifespan, social structure from large image sets quickly and objectively • Acoustic: bat detectors, hydrophones and sonograms record species that are hard to see (bats, cetaceans, insects), giving presence, abundance and activity continuously, day and night	8
05.2	• Data collected continuously / over large areas / at night rather than at a few times and places a researcher can visit • Less disturbance so animals behave naturally; larger sample sizes; objective rather than subjective identification	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402-406; AQA spec 3.1.2.3.6.	
06.1	• A community maintained at a stage before the climax by regular human activity (grazing, mowing, burning, coppicing) that deflects succession • Chalk grassland was produced by centuries of sheep grazing; without it succession to scrub and woodland resumes	2
06.2	• Grazing removes taller / dominant grasses and prevents scrub establishing (5% cover) so small plants are not shaded out • Trampling creates bare ground for wildflower seed germination and warm patches for invertebrates / reptiles • Dung creates small nutrient-rich patches supporting different plants and invertebrates; sheep eat short grasses and some bushes • Mowing removes everything at once including plants grazers avoid, and removes flowers / seeds before setting; burning removes vegetation but also kills invertebrates and allows scrub regrowth; abandoned plot dominated by scrub (55%) so few species	4

Question	Marking guidance	Marks
06.3	• Sheep eat short grass (some breeds eat scrub); ponies avoid flowers but eat bracken; cattle remove long rough grass • Choice depends on the vegetation to be controlled and the plants to be protected	2
06.4	• Removes mature heather bushes and young trees that grazers will not eat, creating open unshaded ground • Heat stimulates dormant heather seeds to germinate so young heather re-establishes	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 47-49; AQA spec 3.1.3.3.	
07.1	• Direct measurements of past conditions cannot be taken, so indirect evidence that is related to the condition must be used	1
07.2	• Age / date of the sample (radioisotope dating) • Past temperatures • Composition of the past atmosphere (eg CO₂ / methane concentrations) • Temperature: more magnesium is incorporated into calcite at higher temperatures	4
07.3	• Movement of continents changed the position of land masses / latitude and therefore the climate and habitats present • Ocean currents redistribute heat and nutrients; changes in currents (eg as continents moved) altered climates, so past conditions cannot be explained without knowing how the systems worked	2
07.4	<i>Any 3 from:</i> • There was no oxygen / ozone layer so high levels of UV radiation reached the land surface; water absorbed the UV and protected organisms in the oceans • Warm temperatures around the vents and a rich mix of chemicals / increasingly complex organic molecules in the sea made biological reactions more likely • The atmosphere contained toxic gases such as ammonia; the early Archaea were anaerobic so did not need oxygen • Water acted as the solvent for reactions / transport	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 6-9; AQA spec 3.1.1.1-3.1.1.3.	
08.1	• Timber: building construction, furniture, tools, telegraph poles, concrete shuttering (oak, beech, mahogany, teak, conifer softwood) • Fuel: fuelwood and charcoal - still the main energy source for most people in LEDCs • Food: fruit, nuts, bushmeat, livestock fodder; crops of forest origin (coffee, cacao, bananas, Brazil nuts); pigs and chickens were forest animals • Fibres: paper (wood pulp cellulose), cotton, viscose / rayon • Medicines: quinine (cinchona), taxol (yew), aspirin (willow); most species not yet researched	5

Question	Marking guidance	Marks
08.2	• Wood is mainly cellulose which is hard to digest, so carbon stored in wood has a long residence time; total forest carbon (vegetation plus soil) is about double that in the atmosphere • Growing trees sequester CO₂ by photosynthesis and release oxygen; forest soils hold more carbon than the trees (especially boreal) • Planting / protecting forests removes CO₂ and reduces the enhanced greenhouse effect	3
08.3	• Highest transpiration of any ecosystem increases downwind rainfall; interception increases evaporation; deep organic soils retain water and delay runoff so river flow is more even and flooding reduced • Low albedo absorbs sunlight; heat stored and released at night reduces day-night temperature extremes; canopy microclimate: lower light, wind, higher humidity	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 360-364; AQA spec 3.5.3.1.	
09.1	• B	1
09.2	• Number all possible sampling sites (eg on a grid map) • Select the sites using a random number generator / table; (regular shape: grid coordinates from random numbers)	2
09.3	• Sites may be chosen for convenience or because they appear to support (or dismiss) the hypothesis • So the sample does not represent the whole area and conclusions are invalid	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 391-393; AQA spec 3.7.1.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Why needed: growing population (to 12 billion), land degradation since the fertile crescent, reliance on artificial fertilisers, pesticides and unsustainable irrigation; limits to new land and imports • Pest control: cultural methods (weeding, mulching, rotation, barrier / companion crops, predator habitats, biological control, sterile males, pheromones), resistant varieties, integrated control, reduced antibiotics • Nutrients: natural processes - nitrogen-fixing legumes, decomposition, recycling organic matter (manure, compost), crop rotation, permaculture, conservation of soil biota and mycorrhizae, reduced artificial fertiliser; phosphate recycling • Energy: natural processes instead of Haber-process fertiliser, low tillage, less machinery, renewable energy, low food miles, reduced methane (cattle diets, rice varieties) • Water: low water-use crops, drip irrigation, reservoirs and aquifer recharge, avoiding saline water; soil: organic matter maintenance, erosion control • Biodiversity: retaining hedgerows, ditches, ponds, woodlands; CWR conservation and seed banks; agri-environment schemes • Social: control of impacts on rural communities, Fairtrade, local food; consumer choices • Evaluation: yields vs sustainability, cost, need for policy support; judgement	9
10.2	• Agricultural systems designed to mimic natural ecosystems - perennial crops, mixed species, recycling of nutrients, minimal external inputs	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 335-337; AQA spec 3.5.1.5.	

Question	Marking guidance	Marks
11.1	Discuss the methods that can be used to conserve biodiversity and evaluate their effectiveness. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Setting priorities: IUCN Red List criteria, EDGE, endemic, keystone, flagship species, population dispersal • Legislation and protocols: Wildlife and Countryside Act, designated areas (SSSI to MCZ), CITES Appendices, IWC, EU CFP, ITTO - successes and limits of enforcement • Captive breeding and release: selection, difficulties, techniques, release site criteria, hard / soft release, monitoring; seed banks • Habitat conservation: creation, design (area, shape, corridors, age structure, colonisation), abiotic and biotic management, plagioclimax maintenance, population control, reintroductions • Habitat case studies (woodland, rainforest, coral reefs, islands, mangroves, Antarctica) illustrating threats and conservation efforts • Monitoring and new technologies informing decisions • Evaluation: examples of success (Iberian lynx, Hawaiian goose, Scilly rats, Yellowstone) and failure (ITTO, tigers); cost, politics, climate change; judgement on the most effective combination Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 26-71; AQA spec 3.1.2.3.	25

Question	Marking guidance	Marks
11.2	Discuss the extent to which the resources and services provided by other species justify the conservation of biodiversity. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Resources: wood, fibres, oils, fuels, food; new food species (morama bean, yeheb, perennial maize, Kernza, bison, eland) and their adaptations • Biomimetics: wing-tip feathers, whale tubercles, shark skin, termite mounds, bird bones, gecko adhesion, burrs, lotus, spider silk • Medicines: poppies, cinchona, willow, yew (taxol), Mexican yam, sponges (AZT), sharks; few species yet studied; physiological research (marsupials, hippo, sponges) • Genetic resources: crop wild relatives, Vavilov centres, gene pools and inbreeding • Ecosystem services: atmospheric regulation, hydrological cycle, biogeochemical cycles, soil maintenance; interspecies relationships - food, pollination (Darwin's orchid), seed dispersal, habitat provision • Evaluation: economic vs intrinsic / moral arguments; services unnoticed because of dynamic equilibrium; unknown future value; justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 10-19; AQA spec 3.1.2.1.	25